

The Machine, Part 1: Paisa aur Machine

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Jo aapke paas ab hai

Ek test, khud ke liye

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THE MACHINE

PART 1 OF 5: PAISA AUR MACHINE

Yeh kitaab paanch chhoti kitaabon mein hai. Yeh pehli hai.

PAANCH PARTS KA NAKSHA

[AAP YAHAN HO]

PART 1	PAISA AUR MACHINE	paisa kyun hai, computer kya hai, sab kuch numbers kaise hai
PART 2	MACHINE SE BAAT	languages, operating system, software banta kaise hai
PART 3	MACHINES KA JAAL	network, address, internet
PART 4	DATA AUR YAADEIN	storage, database, dhoondhna
PART 5	DUNIYA KO SERVE KARNA	scale, cloud, kharcha, poora jod

Yeh kitaab kis liye hai

Aap paisa kamana chahte ho. Aapne duniya ko chaar levels mein dekha hai:

LEVEL 1	INSAAN KI ZAROORAT	bhookh, sehat, gyaan, suraksha
LEVEL 2	BUSINESS	jo zaroorat poori karke paisa leta hai
LEVEL 3	TECHNOLOGY	jo business ko taakat deti hai
LEVEL 4	AI	jo technology ko taakat deti hai

Aap Level 4 pe kaam karna chahte ho, taaki neeche ke saare levels serve kar sako. Lekin Level 4 hawa mein nahi tikta. Woh Level 3 ke upar khada hai. Isliye pehle machine ko samajhna padega. Yeh paanch parts wahi karte hain.

Padhne ke baad aap chaar cheezein kar paoge:

1. Kisi business problem ke liye tech solution ka design khud sochna.
2. AI tools se kaam nikaalna jaise ek engineer nikaalta hai.
3. Koi bhi business dekhkar bataana ki tech se paisa kahan banega.

4. Engineers aur tech news ki baat poori samajhna. Koi aapko bewakoof na bana paaye.

Yeh kitaab kaise likhi gayi hai

Seedhi baat, aapse. Koi lecture nahi. Har cheez ke liye pehle problem, phir naam. Naam hamesha aakhir mein aata hai, kyunki naam pehle aa jaaye toh dimaag sochna band kar deta hai.

Beech beech mein do log milenge: Madhav, ek guru, aur Kabir, ek naujawan jo wahi galtiyan karta hai jo hum sab karte hain. Woh sirf tab aate hain jab koi galat-fehmi todni ho.

Har chapter ka dhancha

CHAPTER	pehle problem, phir samajh, phir naam
NAAM	jo shabd duniya istemaal karti hai
ASLI DUNIYA	ek sachcha example
GALAT-FEHMI	jahan log aksar phislte hain
MAP PE	yeh cheez Level 1-4 mein kahan hai, paisa kahan hai
KHUD DEKHO	5 minute ka kaam, apne phone ya laptop pe (optional)
SOCHNE KE LIYE	ek sawal, jawab seedha neeche

Har chapter ke naam ke saath ek nishaan hai. [SPINE] matlab reedh ki haddi, chhodna mat. [DEPTH] matlab gehrai, pehli baar mein chhod sakte ho, baad mein lautna.

Ek hi instruction

Jaldi mat karo. Ek din mein do-teen chapters kaafi hain. Sawal ka jawab pehle khud socho, phir neeche padho. Jo cheez samajh na aaye, wahan ruko, aage ka sab usi pe khada hoga.

SECTION 0: PAISA, VALUE AUR SOCIETY

Paanch chapters. Yeh section machine ke baare mein nahi hai. Yeh us cheez ke baare mein hai jiske liye aap machine seekh rahe ho: paisa.

Yeh aapke business decision mein kahan aayegi: Jab bhi aap koi idea uthaoge, pehla sawal hoga “paisa kahan se aayega.” Iska jawab hamesha Level 1 se shuru hota hai: kiski zaroorat, kitni badi, kitne log. Yeh section woh sawal poochhna sikhata hai. Iske bina baaki chaar parts sirf gyaan hain, kamaai nahi.

Chapter 0.1 [SPINE]: The village: paisa exist kyun karta hai

Sau logon ka ek gaon socho. Koi paisa nahi hai. Kisan ke paas anaaj hai, mistri ke paas hunar, vaidya ke paas ilaaj, mochi ke paas joote.

Pehla intezaam seedha hai: cheez ke badle cheez. Kisan anaaj dega, mistri chhat banayega. Chal gaya.

Ab ek case dekho. Vaidya ne mochi ka haath theek kiya. Teen mahine baad vaidya ko joote chahiye. Lekin iss baar mochi bilkul theek-thaak hai, usko vaidya se kuch nahi chahiye. Toh vaidya kya kare? Intezaar kare ki mochi bimar pade?

Yahan intezaam toot gaya. Seedha exchange sirf tab chalta hai jab do log ek hi waqt pe ek doosre ka kaam chahte hon. Sau logon ke gaon mein bhi yeh mushkil se hota hai. Yeh pehli toot hai, ise yaad rakho.

Doosra intezaam khud nikaalo. Rukiye, sach mein sochiye, kya karenge?

Bahut log kehte hain: likh lo. Ek register banao. Vaidya ne mochi ka kaam kiya, likh do, baad mein le lega. Achha jawab hai, duniya ne yeh kiya bhi hai. Lekin do naye sawal khade ho jaate hain.

Madhav: Register rakhega kaun?

Kabir: Koi ek aadmi.

Madhav: Aur jis din usne apna naam thoda bada likh diya?

Kabir: Hmm.

Madhav: Aur das hazaar log hon, alag gaon hon, log aate jaate hon. Kitne register?

Kabir: Bahut zyada. Aur kisi ko kisi pe bharosa bhi nahi hoga.

Toh register ka idea sahi hai, uski shakal galat hai. Ab teesra intezaam socho. Aisi koi cheez ho jo mochi le lega, chahe use ilaaj bilkul na chahiye. Woh use kyun lega? Sirf isliye, kyunki use bharosa hai ki baaki sab bhi lenge. Woh cheez uske apne kisi kaam ki

nahi hai. Uska poora value ek hi cheez mein hai: bharosa ki aage badha dunga.

Ab khud nikaalo, us cheez mein kya kya hona chahiye:

1. Sadni nahi chahiye, warna rakhte rakhte khatam.
2. Uthane mein aasan ho, bhaari patthar se kaam nahi chalega.
3. Nakli banana mushkil ho, warna koi bina kuch diye sab le lega.
4. Tukdon mein bat sake, chhoti cheez bhi khareedni hoti hai.
5. Sab log use maanein. Ek bhi na maane toh chain toot jaati hai.

Aapne abhi paisa invent kar diya.

NAAM

Jo cheez aapne nikaali, uska naam **paisa** (money) hai. Pehli toot ka naam economists rakhte hain **double coincidence of wants**: dono ko ek hi waqt ek doosre ki cheez chahiye ho, tabhi seedha exchange chalta hai. Register wale idea ka naam **ledger** hai. Paisa asal mein ek chalta-phirta ledger hai jisme bharosa sabke beech bata hua hai, kisi ek aadmi ke haath mein nahi.

ASLI DUNIYA SE EK EXAMPLE

Aapka UPI dekho. Jab aap kisi ko 50 रुपये bhejte ho, koi cheez haath se haath nahi jaati. Ek register mein likha jaata hai: aapke naam ke aage 50 kam, uske naam ke aage 50 zyada. Woh register banks aur NPCI rakhte hain. Matlab duniya register wale idea pe wapas aa gayi, bas ab register rakhne wala ek aadmi nahi hai, ek system hai jise sab maante hain. Paisa shuru se aakhir tak bharose ka intezaam hai.

YAHAN LOG KYA GALAT SAMAJHTE HAIN

Log sochte hain paisa value HAI. Nahi. Paisa value ka record hai. Note khud kaagaz hai, uska value is bharose mein hai ki agla aadmi use lega. Jis din woh bharosa tootta hai, kaagaz kaagaz reh jaata hai. Aisa hua hai: jin deshon mein logon ka bharosa toota, wahan log boriyon mein note le kar roti khareedne gaye.

MAP PE

Yeh poora chapter Level 1 pe khada hai: insaan ki zaroorat aur unke beech ka len-den. Paisa yahin janamta hai, aur yaad rakho, paisa **SIRF** yahin ghusta hai, jab ek insaan apni zaroorat ke liye deta hai. Level 2, 3, 4 sirf is len-den ko bada, tez aur sasta banate hain. Jo bhi kamaana chahta hai, uski chain kahin na kahin Level 1 ke ek insaan pe khatam hoti hai.

KHUD DEKHO (5 minute)

Apna UPI app kholo. Aakhri 5 transactions dekho. Har ek ke liye poochho: kis zaroorat ke liye paisa gaya? Khaana, safar, gyaan, kuch aur? Har transaction kisi zaroorat pe khatam hoga. Ek bhi aisa nahi milega jo kisi insaan ki zaroorat pe khatam na ho.

SOCHNE KE LIYE

1. (derivation) Agar duniya mein sirf ek hi cheez ka len-den hota, maan lo sirf anaaj, toh kya paisa phir bhi banta?

Jawab: Nahi banta, ya bahut der se banta. Paisa us problem ka hal hai jab bahut alag alag cheezon ka exchange ho aur waqt na mile. Ek hi cheez ho toh seedha exchange kaafi hai, double coincidence wali toot hoti hi nahi. Isse ek gehri baat dikhti hai: paisa tab zaroori hota hai jab society mein kaam bat jaata hai. Jitna zyada kaam ka batwara, utna zaroori paisa. Isi baat pe agla chapter khada hai.

Chapter 0.2 [SPINE]: Same hours, different pay

Do aadmi, ek hi gaon, ek hi umar, dono mehnati.

Ravi mistri hai. Roz 12 ghante kaam karta hai. Har ghar jo woh banata hai, uske haathon se banta hai. Jis din woh kaam nahi karta, us din kamaai zero. Bees saal baad bhi uska din 12 ghante ka hi hai.

Anil ne gaon ke liye paani ki naali ka design banaya. Chhe mahine lage. Uske baad naali ban gayi aur chal padi. Ab paani har ghar pahunchta hai, roz, bina Anil ke haath lagaye. Gaon Anil ko har mahine hissa deta hai. Anil ab doosre gaon ke liye design bana raha hai.

Ravi zyada ghante lagata hai. Anil zyada kamata hai. Kyun?

Pehla jawab jo dimaag deta hai: Anil padha likha hai. Lekin yeh jawab kuch nahi samjhaata. Aisa vaidya bhi padha likha hai jo Ravi jitna hi kamata hai. Asli jawab teen hisson mein hai. Teeno khud nikaalo:

Problem kitni badi hai? Ravi ek ghar ki problem solve karta hai, ek parivaar ke liye. Anil ne poore gaon ki paani ki problem solve ki, sau parivaaron ke liye. Jitne zyada log, jitni gehri zaroorat, utna bada paisa us problem ke peeche khada hai.

Kitne log yeh kar sakte hain? Deewar chunna gaon ke bees log jaante hain. Naali ka design ek hi jaanta tha. Jo cheez kam log kar sakte hain, uska daam upar jaata hai. Yeh mandi ka seedha niyam hai.

Kaam ek baar, faayda kitni baar? Yeh sabse chhupa hua hissa hai. Ravi ki chinaai us din khatam ho jaati hai. Anil ka design ek baar bana aur ab har din kaam kar raha hai, bina Anil ke. Ravi apna waqt bech raha hai. Anil ne ek aisi cheez banayi jo uske sone ke baad bhi chalti hai.

Toh formula yeh nikla:

KAMAAI = PROBLEM KA SIZE x KITNE KAM LOG KAR SAKTE HAIN x KAAM EK BAAR,
FAAYDA BAAR BAAR

Is formula mein dhyaan se dekho: GHANTE KAHIN NAHI HAIN. Isi liye 2 ghante wala arabpati ho sakta hai aur 12 ghante wala tang. Mehnat galat cheez pe lagi ho toh mehnat ka daam nahi milta.

NAAM

Teesre hisse ka naam **leverage** hai: ek baar kaam, baar baar faayda. Doosre hisse ka naam **scarcity** hai: kami. Jo cheez kam hai, mehngi hai. Pehla hissa **market size** kehlata hai. Yeh teen shabd is kitaab mein baar baar aayenge.

ASLI DUNIYA SE EK EXAMPLE

Ek surgeon lo aur ek app banane wala. Surgeon ki skill zyada rare hai, padhaai zyada lambi hai. Phir bhi duniya ke sabse ameer log app aur software wale hain, surgeon nahi. Kyun? Surgeon ka har operation uske apne haathon se hota hai, din mein do ya teen. Leverage ka dial zero pe atka hai. App ek baar banti hai aur ek crore log ek saath use karte hain. WhatsApp jab bika (19 arab dollar), tab uske paas sirf 55 log kaam karte the. 55 log, 40 crore users. Yeh leverage hai.

YAHAN LOG KYA GALAT SAMAJHTE HAIN

Log sochte hain zyada kamaane ka raasta zyada mehnat hai. Formula kehta hai mehnat ka guna tab hi chalta hai jab baaki teen dial upar hon. Sirf ghante badhaane se sirf thakaan badhti hai. Doosri galti: log scarcity ko degree samajhte hain. Scarcity ka matlab certificate nahi, yeh hai ki us waqt, us jagah, woh kaam kitne log kar sakte hain.

MAP PE

Is formula ka teesra dial, leverage, hi asli wajah hai ki Level 3 (technology) itna paisa banata hai. Machine hi woh cheez hai jo ek baar banti hai aur baar baar chalti hai, bina thake. Isi liye aap yeh kitaab padh rahe ho: Level 3 aur 4 pe leverage ka dial sabse upar

tak ghoomta hai.

KHUD DEKHO (5 minute)

Duniya ke 10 sabse ameer logon ki list kholo (search: richest people). Har naam ke aage socho: kya yeh aadmi ghante bech kar yahan pahuncha? Ek bhi nahi milega. Har ek ke paas koi aisi cheez hai jo uske bina chalti hai: company, software, brand, factory.

SOCHNE KE LIYE

1. (derivation) Ek teacher 30 bachchon ko padhaata hai aur mahine ka X kamata hai. Usi gyaan se woh crores kaise kama sakta hai, formula ke teeno dial ghuma kar batao.

Jawab: Size: 30 bachchon ki jagah poore desh ke bachche, online. Scarcity: sab subjects nahi, woh ek cheez jo sirf woh sabse achha samjhaata hai. Leverage: roz class lene ki jagah ek baar recorded course, ya ek kitaab, jo ek baar banti hai aur lakhon baar bikti hai. Physics wallah yahi kahani hai: ek teacher, YouTube pe recorded classes, aaj arabon ki company. Gyaan wahi tha, dial ghoom gaye.

Chapter 0.3 [SPINE]: Society ke chaar levels

Chapter 0.1 ke gaon mein wapas chalo, lekin ab poori duniya socho. Arabon log. Is bheed mein paisa kahan se shuru hota hai aur kahan jaata hai? Ek naksha banate hain, neeche se upar.

Level 1: Insaan ki zaroorat. Bhookh, pyaas, chhat, sehat, gyaan, suraksha, safar, judav, manoranjan, izzat. Yeh list hazaaron saal se lagbhag wahi hai. Yahi poori economy ki zameen hai. Har rupaya jo kahin bhi ghoom raha hai, aakhir mein kisi insaan ki kisi zaroorat se nikla hai.

Level 2: Business. Ek insaan ki zaroorat ek baar poori karna kaam hai. Lakhon logon ki zaroorat baar baar poori karna intezaam maangta hai: log, saamaan, hisaab, bharosa. Us intezaam ka naam business hai. Restaurant bhookh pe khada hai, hospital sehat pe, school gyaan pe. Business zaroorat ko paise mein badalne ki machine hai.

Level 3: Technology. Business ko hamesha teen cheezon ki bhookh hoti hai: zyada log serve karo, kam kharche mein, kam galti ke saath. Jo bhi cheez yeh karti hai, woh technology hai. Tractor kisan ke liye, fridge restaurant ke liye, software har kisi ke liye. Technology business ki taakat badhaane wala level hai.

Level 4: AI. Ab technology khud ek zaroorat wali cheez ban gayi hai: usse sochna, likhna, decide karna, yeh sab abhi tak insaan karte the. Jo machine yeh kaam karne lagi hai, woh technology ki technology hai. AI Level 3 ko wahi deta hai jo Level 3 ne Level 2 ko diya tha: zyada, sasta, kam galti.

Ab sabse zaroori teer kheencho:

LEVEL 4	AI	^	
LEVEL 3	TECHNOLOGY		PAISA UPAR
LEVEL 2	BUSINESS		KI TARAF
LEVEL 1	INSAAN KI ZAROORAT		BEHTA HAI

Paisa SIRF Level 1 pe ghusta hai, jab ek insaan apni zaroorat ke liye deta hai. Level 2 usmein se hissa leta hai. Level 3 ko Level 2 deta hai. Level 4 ko Level 3 deta hai. Upar wale levels ka har rupaya neeche se hoke aaya hai.

Teen baatein check karlo, kyunki saaf naksha inhe chhupa deta hai:

Har zaroorat business se पूरी nahi hoti. Maa bachche ko khila rahi hai, Level 1 seedha serve ho raha hai, koi paisa nahi ghoom raha.

Har business ko technology zyada nahi chahiye. Kuch ko bas thodi si.

Har technology ko AI behtar nahi karta. Paani ki pipe technology hai, usmein sochne wali machine ke liye karne ko kuch nahi.

Yeh naksha jhukav batata hai, kanoon nahi. Aur jahan koi level gayab hai, aksar wahi jagah sabse interesting hoti hai.

NAAM

Is tarah ke naksha ko log **stack** kehte hain: ek cheez doosre ke upar. Aage jab koi kahega “tech stack” ya “AI stack,” toh wahi matlab hai: kaun kis ke upar khada hai, aur kaun kis ko paise deta hai.

ASLI DUNIYA SE EK EXAMPLE

Zomato lo. Level 1: bhokh. Level 2: restaurant, aur khud Zomato ka business. Level 3: app, servers, map, payment system. Level 4: kaunsa khaana kisko dikhaana, delivery ka rasta, demand ka andaaza, yeh AI karta hai. Aap 300 रुपये ka khaana mangwate ho: restaurant hissa leta hai, Zomato hissa leta hai, Zomato apne cloud aur software walon ko deta hai, woh AI chalane walon ko dete hain. Ek bhokh, chaar levels ki kamaai.

YAHAN LOG KYA GALAT SAMAJHTE HAIN

Log sochte hain paisa technology mein hai, isliye tech seekho. Ulta hai. Paisa zaroorat mein hai. Technology sirf use pakadne ka sabse lamba haath hai. Jo aadmi Level 4 ki skill le kar baitha hai lekin kisi Level 1 zaroorat se juda nahi hai, woh kuch nahi kamata.

Yeh kitaab aapko Level 3 aur 4 sikhayegi, lekin har chapter mein hum poochhte rahenge: neechе zaroorat kya hai?

MAP PE

Yeh chapter khud map hai. Ise photo ki tarah dimaag mein rakho. Aage har chapter ke end mein “MAP PE” isi naksha pe ungli rakhega: yeh cheez kis level pe hai, aur paisa is tak kaise pahunchta hai.

KHUD DEKHO (5 minute)

Apne phone ki home screen kholo. Koi 3 apps chuno. Har app ke liye likho: kaunsi Level 1 zaroorat? Kaunsa Level 2 business? WhatsApp: judav ki zaroorat, Meta ka business. Aisa app dhoondho jo kisi zaroorat pe khada NA ho. Nahi milega.

SOCHNE KE LIYE

1. (derivation) AI companies (Level 4) arabon kama rahi hain. Lekin paisa sirf Level 1 pe ghusta hai. Toh OpenAI ka paisa kis raaste se, kin levels se hoke, aata hai? Chain likho.

Jawab: Ek chain aisi ho sakti hai: aap bhookh mein khaana order karte ho (L1 se paisa ghusa). Zomato kamata hai (L2). Zomato apne software aur cloud pe kharch karta hai (L3). Woh software banane wali team ChatGPT ka subscription leti hai taaki code tez likhe (L4). Har AI ka rupaya aise kisi na kisi chain se neechе ke levels se hoke aaya hai. Iska ulta bhi sach hai: agar koi AI product kisi bhi chain se Level 1 tak nahi pahunchta, toh uske paas aane wala paisa hai hi nahi.

Chapter 0.4 [SPINE]: Technology level dikhta kyun nahi

Aaj raat aap phone se khaana order karte ho. Aapko kya dikhta hai? Ek button: “Place Order.” Bas.

Ab ginte hain ki us ek button ke peeche kya chal raha hai. Aapka phone ek message bana kar hawa mein bhejta hai. Woh message shehar paar ke kisi building mein rakhi machine tak jaata hai. Woh machine aapka naam, pata, order, paise ka hisaab jaanchti hai. Bank ki machine se baat karke 300 रुपये idhar se udhar karti hai. Restaurant ki screen pe order chamakta hai. Ek aur machine sochti hai ki kaunsa delivery wala kahan hai aur kise bhejna sasta padega. Aapke naksha pe woh chalta hua dikhta hai.

Yeh sab 5 second mein hua. Aur aapko inmein se KUCH BHI nahi dikha.

Yeh koi galti nahi hai. Yeh design hai. Kyun? Do wajah, dono paise ki:

Pehli: Aap bhookhe ho. Aap khaane ke paise de rahe ho, machine dekhne ke nahi. Har extra cheez jo aapko dikhegi, aapko rokegi, sochne pe majboor karegi, aur ruka hua aadmi order cancel karta hai. Business ke liye har dikhai deti hui machinery paisa girne ka khatra hai. Isliye jo technology jitni achhi hoti hai, utni gayab hoti hai.

Doosri: Jo cheez gayab hai, use koi seekh nahi sakta. Jo seekh nahi sakta, woh bana nahi sakta. Toh dikhne wala hissa (button) sasta ho gaya, sab bana lete hain. Na dikhne wala hissa (peeche ka intezaam) mehnga raha, kyunki kam log jaante hain. Chapter 0.2 ka scarcity wala dial yaad hai? Technology ki scarcity uski invisibility se aati hai.

Ab is kitaab ka maqsad ek line mein: **jo hissa business ne jaan boojh ke chhupaya hai, hum use khol ke dekhenge.** Kyunki paisa wahi hissa banata hai.

NAAM

Chhupaane ke is kaam ka naam engineers rakhte hain **abstraction**: andar ki complexity dhak kar upar ek seedha sa handle dena. Button abstraction hai. Steering wheel abstraction hai (engine chhupa hai). UPI ka QR code abstraction hai (banks ka

poora jaal chhupa hai). Yeh shabd is kitaab mein sau baar aayega, kyunki poori technology abstraction ke upar abstraction ki seedhi hai.

ASLI DUNIYA SE EK EXAMPLE

Jio ne 2016 mein muft data diya aur desh badal gaya. Logon ko kya dikha? Ek sasta SIM. Kya nahi dikha? Samundar ke neeche padi fibre ki taarein, har shehar mein towers, unke beech ka intezaam, is sab pe lagaya gaya 2.5 lakh crore rupaya. Level 3 itna bada tha ki desh ki sabse badi company ban gaya, aur aam aadmi ne use kabhi dekha hi nahi. Yehi baat aapke seekhne ke liye sabse badi khabar hai: sabse bade paise wale kaam aankhon se ojhal hain.

YAHAN LOG KYA GALAT SAMAJHTE HAIN

Log “technology” sunte hi gadget sochte hain: phone, laptop, robot. Gadget technology ka sirf woh kona hai jo bikne ke liye dikhaya jaata hai. Asli technology intezaam hai: numbers rakhne ka, bhejne ka, jaanchne ka, decide karne ka. Isi galat-fehmi ki wajah se log sochte hain “tech seekhna matlab coding seekhna.” Coding ek auzaar hai. Samajhna yeh hai ki poora intezaam kaise khada hota hai.

MAP PE

Rupaye ka rasta dekho: aapke 300 रुपये mein se restaurant ko ~200, Zomato ko ~100 mile. Us 100 mein se Zomato cloud, payment gateway, map service, SMS bhejne walon ko deta hai. Yeh sab Level 3 ke andar ke business hain jinka naam aapne kabhi nahi suna: Razorpay, AWS, Twilio. Na dikhne wale level ke andar bhi poore bazaar hain. Jahan aankh nahi jaati, wahan competition bhi kam hoti hai.

KHUD DEKHO (5 minute)

Koi bhi online payment karo (ya purani payment yaad karo). Ab kaagaz pe likho: is 5 second mein kitni alag companies ne kaam kiya hoga? Ginti karo: aapka phone banane wali, network wali, app wali, bank, payment gateway, receiver ka bank, NPCI. Ek tap, saat se zyada companies. Har ek ko us tap se paisa milta hai.

SOCHNE KE LIYE

1. (derivation) Agar koi business apni technology chhupaane ki jagah poori dikhaane lage (har order pe: “ab server X se baat ho rahi hai...”), toh uska kya hoga? Aur is jawab se yeh nikaalo ki engineer ki tankhwah zyada kyun hoti hai.

Jawab: Users bhaag jaayenge, kyunki unhe zaroorat poori karni thi, machine nahi samajhni thi. Business dikhaane ke liye nahi, chhupaane ke liye paise deta hai. Aur jo cheez sabse chhupi hui hai, use samajhne wale sabse kam log hain. Kam log = scarcity = unchi tankhwah. Engineer ko is baat ke paise milte hain ki woh us kamre mein kaam kar sakta hai jisme baaki duniya ghus hi nahi sakti. Aap yeh kitaab padh ke usi kamre ki chaabi bana rahe ho.

Chapter 0.5 [SPINE]: Toh ab hume kya kholna padega

Chaar chapters mein aapke paas ab yeh hai:

1. Paisa bharose ka intezaam hai, aur SIRF Level 1 pe ghusta hai. (0.1)
2. Kamaai = size x scarcity x leverage. Ghante formula mein nahi hain. (0.2)
3. Duniya chaar levels ka stack hai, paisa neeche se upar behta hai. (0.3)
4. Level 3 jaan boojh ke chhupa hai, aur chhupa hona hi uska daam hai. (0.4)

Ab ek imaandaar sawal: aap Level 1 aur 2 pehle se samajhte ho. Zaroorat kya hoti hai, business kaise chalta hai, yeh aapko dikhta hai. Jo nahi dikhta, woh Level 3 aur 4 hai. Aur aapka plan Level 4 pe kaam karna hai.

Yahan ek chhota samvaad zaroori hai, kyunki yahan sab phislte hain.

Kabir: Toh main seedha AI kyun na seekhun? Mujhe wahi chahiye.

Madhav: AI kis pe chalta hai?

Kabir: Computer pe.

Madhav: Aur tumhe computer samajh nahi aata. Toh AI tumhare liye kya hoga?

Kabir: ...jaadu.

Madhav: Aur jaadu ke saath tum kya kar sakte ho?

Kabir: Dekh sakta hoon. Use kar sakta hoon, thoda.

Madhav: Design nahi kar sakte. Bech nahi sakte. Sudhaar nahi sakte. Jab woh toote toh bata nahi sakte kya toota. Jaadu ka darshak ban sakte ho, jaadugar nahi.

Isliye raasta neeche se hai. Level 4 Level 3 pe khada hai, toh pehle Level 3: machine. Yeh Part 1 se Part 5 tak ka safar hai:

PART 1 (yeh wali)	machine kya hai, sab kuch numbers kaise hai
PART 2	machine ko batana ki kya karna hai: software
PART 3	machines ka aapas mein baat karna: internet
PART 4	yaad rakhna: data, database
PART 5	lakhon logon ko serve karna: scale, cloud, kharcha

Iske baad Book 2 aati hai: AI. Tab woh jaadu nahi lagegi, machine lagegi.

Aur ek daava likh ke rakho, hum ise Part 5 ke end mein test karenge:

Daava: Machine samajhne ke baad aap kisi bhi business ko dekh kar bata paoge ki usme technology se paisa kahan banega, kya banana sasta hai, kya mehnga, aur AI usme kahan ghusega.

YAHAN LOG KYA GALAT SAMAJHTE HAIN

”Basics baad mein dekh lunga, pehle jaldi wala raasta.” Aisa raasta hota toh sab us pe hote. Jo log AI tools ko sabse achha chalte hain, woh wahi hain jo jaante hain ki andar kya ho raha hai, isliye woh sahi cheez maang paate hain aur galat jawab pakad lete hain. Machine ki samajh hi AI ki steering hai.

MAP PE

Aap is waqt map pe khud ko rakh rahe ho: Level 1-2 ki samajh aapke paas hai, Level 3 ka darwaza yeh kitaab hai, Level 4 agli kitaab. Jis din aapke paas chaaron levels ki samajh hogi, aap us chhoti si bheed mein honge jo poori chain dekh sakti hai. Chain dekhne wala hi chain design kar sakta hai.

KHUD DEKHO (5 minute)

Ek kaagaz pe apne haath se yeh likho aur photo kheench lo: “Main yeh isliye seekh raha hoon taaki _____.” Apne shabdon mein bharo. Part 5 ke aakhri chapter mein yeh line wapas maangi jaayegi, tab dekhna ki jawab badla ya wahi hai.

SOCHNE KE LIYE

1. (derivation) Kabir wala tark ulta karo: ek aadmi ne SIRF Level 3 seekha, machine ka poora expert, lekin Level 1-2 (zaroorat, business) kabhi nahi dekha. Uski kamaai ka kya hoga?

Jawab: Woh kamayega, lekin hamesha kisi aur ke liye. Tankhwah milegi, kyunki uski skill scarce hai. Lekin size aur leverage ke dial koi aur ghumayega: founder, jo jaanta hai kis zaroorat pe kya banana hai. Duniya ke lakhon behtareen engineers tankhwah pe hain aur unke founders arabpati hain. Farq skill ka nahi, chain ki poori samajh ka hai. Isliye is kitaab mein har chapter money line ke saath aata hai: aapko dono aankhein chahiye, machine wali bhi, paisa wali bhi.

SECTION 1: PEHLA SAWAL

Chhe chapters. Yahan se machine khulti hai: switch se le kar is baat tak ki ek hi machine sab kuch kyun kar sakti hai.

Yeh aapke business decision mein kahan aayegi: Jab aap kisi engineer ya AI se product banwaoge, toh “kya possible hai, kya nahi, kya sasta hai, kya mehnga” ka andaaza isi section ki samajh se aayega. Jo founder machine ko jaadu samajhta hai, woh galat cheez maangta hai, galat daam pe, galat aadmi se. Jo samajhta hai, woh theek wahi maangta hai jo machine de sakti hai.

Chapter 1.1 [SPINE]: Ek switch dabane se kya hota hai

Deewar ka switch dabao. Bulb jala. Yeh poori kitaab isi ek ghatna ke andar chhupi hai, toh ise dheere se kholte hain.

Bijli ghar mein taaron ke raaste aati hai. Taar ek bandh raasta hai: power station se chal kar, bulb se hote hue, wapas. Jab tak raasta poora juda hua hai, bijli behti hai aur bulb jalta hai. Switch us raaste ki ek kadi hai jo tootti aur judti hai. Daba diya: raasta juda, bijli behi, roshni. Wapas dabaya: raasta toota, andhera.

Toh switch asal mein kya hai? **Ek sawal ka jawab jo do mein se ek ho sakta hai: juda hai, ya toota hai. Haan, ya na. 1, ya 0.**

Ab teen chhalaang lagao, har ek chhoti hai:

Pehli: Switch ungli se dabta hai. Kya bijli se dab sakta hai? Haan, aisa switch banta hai jise doosri taar ka current daba deta hai. Iska matlab dhyaan se suno: **ek switch doosre switch ko daba sakta hai.** Ungli ki zaroorat sirf shuru mein hai.

Doosri: Agar ek switch doosre ko daba sakta hai, toh hazaar switch mila kar ek chain ban sakti hai: yeh dabega toh woh dabega, woh dabega toh yeh khulega. Chain design karne ka matlab hai: pehle se tay karna ki kya hone pe kya hoga. Yeh “pehle se tay karna” aage bahut bada hone wala hai.

Teesri: Aaj ke switch mein koi hiltta hua purza nahi hai. Woh retili dhaatu (silicon) ka ek tukda hai jo bijli ke ishaare pe raasta kholta band karta hai. Na ghisna, na awaaz, aur size? Aapke phone ke andar wali chip mein aise 15 arab (billion) switch hain. Nakhoon ke size mein.

Toh yeh kitaab jis machine ko khol rahi hai, woh yeh hai: **arabon switch, ek doosre ko dabate hue, ek design ki hui chain mein.** Aapka har tap, har message, har video isi ka natija hai. Agle chapters mein hum dekhte hain ki on/off se ginti, photo, aur soch kaise banti hai.

NAAM

Bijli se dabne wale switch ka naam **transistor** hai. Yeh shayad insaan ki sabse zaroori invention hai, aur duniya isse har saal kharabon ki ginti mein banati hai. Bandh raaste ka naam **circuit** hai. Haan/na wale jawab ka naam agla chapter dega.

ASLI DUNIYA SE EK EXAMPLE

Aapke phone ka torch button dabao. Ungli ne sheeshe ko chhua, sheeshe ke neeche ke sensor ne ek chhota sa current bheja, us current ne chain shuru ki, aur chain ke aakhir mein ek transistor ne LED ka raasta jod diya. Roshni. Aapne deewar wala switch hi dabaya, bas beech mein arabon switch the. Farq sirf ginti ka hai, idea wahi hai.

YAHAN LOG KYA GALAT SAMAJHTE HAIN

Log sochte hain computer koi ajeeb, samajh ke bahar ki cheez hai. Sach yeh hai ki uska har hissa utna hi seedha hai jitna deewar ka switch. Jo cheez dimaag ko chakrati hai woh GINTI hai: arabon switch, arab baar prati second. Lekin ginti se darna aisa hai jaise samundar se darna kyunki usmein boondein bahut hain. Boond samajh lo, samundar samajh aayega.

MAP PE

Level 3 ki poori imaat is ek purze pe khadi hai, aur transistor banane ka business duniya ke sabse bade businesses mein hai. Ek company, TSMC (Taiwan), duniya ke sabse behtareen transistor banati hai, aur Apple se le kar Nvidia tak sab uske darwaaze pe line mein hain. Ek cheez jo koi aur nahi bana sakta = scarcity ka dial poora upar. Ek factory jo ek baar lagti hai aur arabon chips nikaalti hai = leverage bhi poora upar. Formula kaam kar raha hai.

KHUD DEKHO (5 minute)

Ghar ka koi switch dabate waqt yeh sochna: main ek sawal ka jawab set kar raha hoon, haan ya na. Phir phone ka torch on karna aur khud se kehna: abhi maine arabon switchon ki chain ke sire pe ungli rakhi. Dono ek hi cheez hain. Yeh mehsoos karna hi is

chapter ka asli kaam hai.

SOCHNE KE LIYE

1. (derivation) “Ek switch doosre switch ko daba sakta hai.” Agar yeh possible NA hota, ungli hamesha zaroori hoti, toh computer ka kya hota?

Jawab: Computer ban hi nahi paata. Har kadam pe insaan chahiye hota, toh machine ki speed insaan ki speed pe atak jaati: second mein arab kadam ki jagah second mein ek kadam. Aur leverage zero ho jaata: machine “ek baar bana ke chhod do, chalti rahegi” wali cheez na hoti. Poora Level 3 is ek baat pe khada hai ki chain ek baar design karo, phir woh bina insaan ke chalti hai. Anil ki naali yaad karo: yeh wahi hai, bas paani ki jagah bijli behti hai.

Chapter 1.2 [SPINE]: Yeh maan kar chal raha hoon

Har kitaab kuch cheezein maan kar chalti hai aur batati nahi. Phir reader kahin atakta hai aur use lagta hai woh kamzor hai. Galat. Kitaab ne farz kiya tha, yeh kitaab farz nahi karegi. Yeh raha poora list jo main maan raha hoon. Jo pata hai, use tick karo, jo nahi, uska ek line ka jawab yahin hai.

Bijli kya hai? Dhaatu ke andar chhote kan (electrons) hote hain jo dhakka milne pe ek taraf sarakte hain. Us sarakne ka naam current hai. Paani ki pipe mein paani jaisa soch lo, kaam chalega. (Analogy hai, poori sachchai nahi, lekin is kitaab ke liye kaafi hai.)

Taar current kyun le jaati hai aur plastic kyun nahi? Kuch cheezon mein kan aasani se sarakte hain (dhaatu: conductor), kuch mein bilkul nahi (plastic, rubber: insulator). Isliye taar dhaatu ki hai aur uska khol plastic ka: andar raasta, bahar deewar.

Battery kya karti hai? Dhakka deti hai. Uske do sire hote hain, aur woh kanon ko ek sire se doosre tak dhakelti hai. Jab tak raasta juda hai, dhakka chalta hai.

”Signal” kya hai? Current ka hona ya na hona, bas. Jab hum aage kahein “phone ne signal bheja,” matlab kisi taar ya hawa mein current/ lehar ka pattern gaya: hai, nahi hai, hai, hai, nahi hai. Isse zyada kuch nahi.

Chip kya hai? Silicon ke chhote tukde pe chhappe hue arabon transistor aur unke beech ki taarein, ek dibbi mein band. Computer ke andar ki hari patti (circuit board) pe yahi dibbiyan lagi hoti hain.

Bas. Is kitaab ko bijli ke baare mein isse zyada kuch nahi chahiye. Agar upar ki paanch baatein theek lagti hain, toh aapke paas poori kitaab ke liye kaafi physics hai.

YAHAN LOG KYA GALAT SAMAJHTE HAIN

Log sochte hain technology samajhne ke liye pehle physics ka pahad chadhna padega, phir maths ka. Nahi. Har level apne neeche wale level ka sirf ek chhota sa hissa use karta hai. Computer banane wale ko bhi electron ki poori physics nahi chahiye, use bas

itna chahiye: switch kholta hai, band karta hai, bharosemand hai. Yeh baat aage har jagah lagegi: **har level, neeche wale ka sirf handle pakadta hai, poora engine nahi.** (Yeh abstraction hai, Chapter 0.4 wali.)

MAP PE

Kaun kamata hai yahan? Jo log yeh neeche wala level poora jaante hain: electrical engineers, chip designers. Unki duniya is kitaab ke bahar hai, aur wahan bhi arabon ka bazaar hai (power grids, batteries, solar). Hum us darwaaze ke saamne se guzar rahe hain, ghus nahi rahe. Yaad bas itna rakho ki woh darwaza hai, aur uske andar bhi Level 3 hi hai: kisi business ki taakat badhaana.

KHUD DEKHO (5 minute)

Apna phone charger dekho. Us pe chhota sa likha hoga: “5V 2A” jaisa kuch. V dhakke ka naap hai (kitni zor se), A bahaav ka naap hai (kitne kan prati second). Ab charger ki taar dekho: andar dhaatu, bahar plastic. Conductor, insulator. Aapke haath mein poora Chapter 1.2 hai.

SOCHNE KE LIYE

1. (derivation) Paani wali analogy kahan toot sakti hai? Ek jagah socho jahan bijli paani jaisi NAHI hai.

Jawab: Kai jagah. Sabse badi: paani pipe se girta hai toh farsh geela hota hai, lekin current ko poora juda hua raasta chahiye, bina wapas jaane ke raaste ke current behta hi nahi. Isliye chidiya ek taar pe baith sakti hai (raasta poora nahi hua), lekin do taar chhoo le toh raasta poora ho jaata hai. Analogy auzaar hai, sachchai nahi. Yeh sabak aage AI tak jaayega: har model, har analogy kahin na kahin tootti hai, aur expert wohi hai jo jaanta hai kahan.

Chapter 1.3 [SPINE]: Computer actually hai kya

Ab tak ke tukde jodo: switch haan/na rakhta hai, switch switch ko dabata hai, chain pehle se design hoti hai. Toh yeh poori cheez milkar karti kya hai?

Ek kaam se shuru karo jo aap khud karte ho: $47 + 38$ jodna. Aap kya karte ho? Aapke paas ek TAREEKA hai, bachpan mein rata hua: pehle ikai jodo, $7 + 8 = 15$, 5 likho, 1 haath mein. Phir dahai jodo, haath wala 1 bhi. Jawab 85. Dhyaan do: aapne SOCHA nahi, aapne ek rata hua tareeka CHALAYA. Koi bhi do numbers do, tareeka wahi rahega.

Ab poochho: kya switchon ki chain se aisa tareeka chal sakta hai? Haan. Numbers haan/na mein rakhe ja sakte hain (agla chapter). Jodne ka tareeka switchon ki chain mein pehle se design kiya ja sakta hai (Chapter 1.5). Toh ek dibba ban sakta hai: is taraf se do numbers ghusao, us taraf se jod nikle. Bina kisi insaan ke.

Aur ab asli chhalaang. Agar jodne ka dibba ban sakta hai, toh: compare karne ka dibba (kaunsa bada?), copy karne ka dibba, “agar yeh toh woh” ka dibba. Aur in dibbon ko line mein laga do, ek ke baad ek, toh lambi recipe chal sakti hai: “pehle jodo, phir compare karo, agar bada hai toh yeh copy karo...”

Toh computer yeh hai: ek machine jo numbers pe, pehle se likhi recipe, bina thake, arab kadam prati second ki raftaar se chalati hai.

Teen shabd is definition mein dhyaan se dekho:

Numbers pe. Computer sirf numbers jaanta hai. Photo, gaana, message, sab pehle number banenge (Section 2 ka kaam).

Pehle se likhi. Computer ke har kadam ke peeche kisi insaan ka pehle se liya gaya faisla hai. Machine faisla nahi karti, faisla CHALATI hai.

Bina thake. Yehi iski poori keemat hai. Recipe ek baar likho, machine arab baar chalayegi. Leverage, shuddh roop mein.

NAAM

Recipe ka naam **program** hai. Recipe likhne ka naam **programming**. Saari recipes milke **software** kehlati hain, aur chhoone laayak machine **hardware**. Jodne-compare karne wala mukhya dibba **processor** ya **CPU** kehlati hai.

ASLI DUNIYA SE EK EXAMPLE

Bank ka clerk 1960 mein interest ginta tha: har khaate pe wahi kadam, din bhar, galtiyan ke saath. Aaj wahi recipe software mein likhi hai aur raat ko 2 baje har khaate pe ek saath chalti hai, karodon baar, zero thakaan. Clerk ka TAREEKA hi machine mein gaya. Insaan ka kaam recipe likhna reh gaya, chalana machine le gayi. Jahan jahan duniya mein koi “rata hua tareeka” baar baar chal raha hai, wahan wahan computer ghusega. Yeh line business dhoondhne ka auzaar hai, sambhaal ke rakho.

YAHAN LOG KYA GALAT SAMAJHTE HAIN

“Computer sochta hai.” Nahi. Woh recipe chalata hai. Jab computer “galti” karta hai, woh recipe ki galti hai, kisi insaan ki likhi hui. Jab computer “decide” karta hai, woh insaan ka pehle se likha faisla hai. Yeh baat AI tak bhi jaayegi, wahan recipe ajeeb tarike se likhi jaati hai, lekin machine wahi hai. Jo aadmi yeh samajh gaya, woh machine se kabhi nahi darega aur us pe kabhi andha bharosa bhi nahi karega.

MAP PE

Is chapter ne Level 3 ko do hisson mein kaat diya: hardware aur software. Dono ke bazaar alag hain. Hardware mein paisa factory aur scarcity se aata hai (TSMC, Samsung, Nvidia). Software mein paisa LIKHNE ki laagat ek baar lagne se aata hai: recipe ek baar likhi, ab har copy muft. Isliye duniya ki sabse moti kamaai wali companies software wali hain: Microsoft ka maal banane ka kharcha lagbhag zero hai. Aap jo seekh rahe ho, woh isi taraf ka darwaza hai.

KHUD DEKHO (5 minute)

Phone ka calculator kholo. $47 + 38$ karo. Ab khud se poochho: iss glass ke neeche kya hua? Aapke tap numbers bane, numbers CPU ke jodne wale dibbe mein gaye, jawab ke numbers screen ke pixels bane. Poora Chapter 1.3 ek second mein chala. Aur yeh calculator recipe kisi ne EK BAAR likhi thi, ab arabon phones pe muft chal rahi hai.

SOCHNE KE LIYE

1. (derivation) “Machine faisla nahi karti, faisla chalati hai.” Toh jab bank ka software aapka loan reject karta hai, asli faisla kisne kiya? Aur is jawab se business ki ek zimmedari nikaalo.

Jawab: Faisla us insaan ne kiya jisne recipe likhi ya likhwayi: “agar income itni se kam, toh na.” Machine ne sirf use bina thake chalaya. Zimmedari: jab business apna faisla recipe mein daal deta hai, toh galti bhi leverage pa jaati hai, ek galat नियम lakhon logon pe ek saath chalta hai. Isliye recipe likhne wale ki keemat bhi zyada hai aur zimmedari bhi. AI ke zamaane mein yeh sawal aur bada hoga, Book 2 mein milega.

Chapter 1.4 [SPINE]: Switches numbers kaise rakhte hain

Ek switch do baatein keh sakta hai: 0 ya 1. Lekin duniya ke numbers toh 0 se 9 tak ke ankon se bante hain, aur bade numbers, jaise 300 रुपये ya 15 arab, unka kya?

Khud nikaalte hain. Ek switch: do haalat (0, 1). Do switch: chaar haalat (00, 01, 10, 11). Teen switch: aath haalat. Har naya switch haalat ki ginti DUGNI kar deta hai, kyunki purani har haalat ab do roop mein ho sakti hai: naya switch 0 ke saath, ya 1 ke saath.

Ab in haalat ko naam de do, bas itna hi karna hai:

00 ko bolo "zero"	01 ko bolo "ek"
10 ko bolo "do"	11 ko bolo "teen"

Ruko, yeh toh wahi hai jo hum das ungliyon ke saath karte hain! Hum 9 ke baad naya anka shuru karte hain (10), machine 1 ke baad naya switch shuru karti hai (10). Hamara har anka das guna badhta hai, machine ka har switch do guna. Bas itna hi farq hai. Ginti ka idea wahi hai, sirf ungliyaan 10 ki jagah 2 hain.

Toh kitne switch chahiye kaam ke numbers ke liye?

8 switch	=	256 haalat	(0 se 255)
16 switch	=	65,536	
32 switch	=	429 crore	
64 switch	=	itna bada ki duniya ke saare रुपये, saare phone, saari files gin lo, khatam nahi hota	

Aapke phone ka processor ek saath 64 switchon ke jhund pe kaam karta hai. Isliye "64-bit phone" likha hota hai dibbe pe. Ab aap jaante ho us line ka matlab, duniya ke 99% log nahi jaante.

Ek aakhri sawal khud se poochho: machine 2 pe kyun ruki, 10 haalat wala switch kyun nahi banaya? Ban sakta tha. Lekin bijli ka current kabhi thoda upar neeche hota rehta hai (noise). Do haalat mein farq itna bada hai (poora on, poora off) ki chhoti moti hilna-dulna se galti nahi hoti. Das haalat paas paas hoti, aur har halki kampan galat anka ban

jaati. **2 chuna gaya kyunki 2 sabse bharosemand hai.** Machine ki har design mein yeh sawal milega: tez chahiye, sasta chahiye, ya bharosemand chahiye? Yahan bharosa jeeta.

NAAM

Ek switch ki haalat ka naam **bit** hai (binary digit). Aath bit ke jhund ka naam **byte**. Do-guna wali ginti ka naam **binary**. Yeh teen shabd ab aapke hain, aur poori digital duniya inhi se napi jaati hai.

ASLI DUNIYA SE EK EXAMPLE

Aapka bank balance kahin kisi machine mein 64 switchon ki ek qataar hai. UPI se 50 रुपये bheje toh us qataar ke kuch switch palte, aur kisi doosri machine mein doosri qataar ke switch palte. Poori Indian economy, har khaata, har lena-dena, on/off ki qataaron mein rakhi hai. Isliye us qataar ki hifaazat (Part 3 mein encryption) itna bada business hai: switchon mein rakha paisa asli paisa hai.

YAHAN LOG KYA GALAT SAMAJHTE HAIN

Log binary dekh ke sochte hain machine koi alag, ajnabi bhasha bolti hai jo insaan ke bas ki nahi. Aapne abhi 5 minute mein poora idea nikaal liya, kyunki naya kuch tha hi nahi: yeh wahi ginti hai jo aap bachpan se karte ho, chhoti ungliyon pe. Technology mein “mushkil” aksar “anjaan” ka doosra naam hota hai.

MAP PE

Rupaye ka rasta: har cheez jo aap store karte ho, bits mein napti hai, aur bits rakhne ki jagah ka kiraya lagta hai. Google aapko 15 GB muft deta hai, aur uske upar mahina charge karta hai. Woh kiraya arabon ka business hai (Google One, iCloud). Aage Chapter 2.1 mein hum yeh naap poora seekhenge, kyunki jo cheez napti hai, wahi bikti hai.

KHUD DEKHO (5 minute)

Ek haath ki paanch ungliyan lo. Band ungli 0, khuli 1. Angootha se shuru karo: sirf angootha khula = 1, sirf doosri ungli = 2, dono = 3. Aise ginte jao. Paanch ungliyon pe aap 31 tak gin sakte ho. Jab yeh kar loge, aapne binary ko haath se chala liya, ab woh kabhi nahi bhoolegi.

SOCHNE KE LIYE

1. (derivation) 8 switch 256 haalat dete hain. Ek purani game console “8-bit” thi. Uske screen pe ek rang chunne ke liye 8 bit the. Us console pe kitne rang possible the, aur is se yeh nikaalo ki purane games waise kyun dikhte the?

Jawab: 256 rang, bas. Isliye purane games ke rang saaf saaf tukdon mein dikhte hain, godhuli ka narm dhalta rang unme ban hi nahi sakta tha. Aaj ke screen har pixel ke liye 24+ bit rakhte hain, 1.6 crore rang, isliye photo asli lagti hai. Sabak: bits ki ginti seedha tay karti hai ki machine kitni bareeki dekh sakti hai. Yeh niyam AI tak jaayega: wahan bhi bareeki bits mein bikti hai.

Chapter 1.5 [DEPTH]: Switches decide kaise karte hain

(Yeh DEPTH chapter hai. Pehli baar mein bhaari lage toh aage badho, Part 1 khatam karke lautna. Lekin laut zaroor aana, kyunki yahin pe “machine sochti hai” wala jaadu poora tootta hai.)

Chapter 1.3 mein humne maana tha ki “jodne ka dibba” aur “compare karne ka dibba” ban sakte hain. Ab banate hain. Sirf teen chhote purze chahiye, teeno switchon se bante hain:

AND: do raaste ek ke baad ek (line mein). Bijli tabhi paar hogi jab DONO switch on hon. Matlab: “dono haan, toh haan.”

OR: do raaste side by side (samanantar). Bijli paar hogi agar KOI BHI ek on ho. Matlab: “koi bhi haan, toh haan.”

NOT: ulta karne wala. Input 1 toh output 0, input 0 toh output 1.

Bas. Yeh teen hi hain. Ab inse kaam karwao.

Compare banao: “Kya A aur B alag hain?” Socho: alag ka matlab hai (A haan, B na) YA (A na, B haan). Yeh toh AND, OR, NOT ka jod hai: $(A \text{ AND } (\text{NOT } B)) \text{ OR } ((\text{NOT } A) \text{ AND } B)$. Ho gaya. Machine ab “alag hai ya nahi” bata sakti hai. Aur “barabar hai?” uska NOT hai.

Jodna banao: ek bit jodo: $0+0=0$, $0+1=1$, $1+0=1$, $1+1=10$ (do bits ka jawab: “0 likho, 1 haath mein”). Dekho, jawab ka pehla hissa “alag hain?” wala purza hai, aur haath wala hissa AND hai (dono 1 hon tabhi haath mein jaata hai). School wali jod ki recipe, switchon mein utar gayi. Aise 64 jodo, aur 64-bit numbers ka jod tayyar.

”Agar” banao: “AGAR yeh 1 hai TOH is taar ka signal aage jaane do.” Yeh sirf ek AND hai: signal AND shart. Recipe ka har “agar aisa toh waisa” aakhir mein inhi AND/OR/NOT ki chain hai.

Ab poora minaar neeche se upar dekho:

```
transistor  -> AND/OR/NOT  -> jod/compare/agar  -> recipe ke kadam  
            -> poora program -> WhatsApp
```

Har manzil apne neeche wali manzil ke purzon se bani hai. Kahin bhi, kisi bhi manzil pe, “soch” naam ka koi purza nahi hai. Sirf switch hain, design mein jade hue.

NAAM

In purzon ka naam **logic gates** hai. AND gate, OR gate, NOT gate. “Logic” isliye ki yeh haan/na wale tarkon ko bijli mein utaarte hain. Aapke phone ki chip in gates ke arabon jodon se bani hai, aur gates transistors se.

ASLI DUNIYA SE EK EXAMPLE

ATM ka faisla: “paise do AGAR pin sahi hai AND balance kaafi hai AND aaj ki limit baaki hai.” Teen shartein, ek AND chain. Jis din aap ATM se paisa nikaalte ho, ek AND gate ka jhund aapke haq mein 1 bolta hai. Bank ka poora risk intezaam aise hi gates ki chains mein likha hai.

YAHAN LOG KYA GALAT SAMAJHTE HAIN

“Machine decide karti hai, matlab uske andar kahin koi chhota dimaag hai.” Ab aap jaante ho decide ka matlab kya hai: pehle se design ki hui AND/OR/NOT ki chain, jisme har possible input ka output PEHLE SE tay hai. Machine ke paas surprise nahi hota. Jab machine ka jawab aapko surprise kare, toh matlab aap design nahi jaante, machine kuch naya nahi soch rahi.

MAP PE

Kaun kamata hai: chip design karne wale. Gates ko jodkar tez, chhoti, kam bijli wali chain banana duniya ki sabse scarce skills mein hai. Nvidia isi khel mein duniya ki sabse keemti companies mein pahunchi: unke gates ka intezaam AI ke kaam ke liye sabse tez nikla. Jo cheez aapne aaj 15 minute mein samjhi, usi ki behtareen jamawat kharabon dollar ki company banati hai. Farq gehraai ka hai, idea ka nahi.

KHUD DEKHO (5 minute)

Kaagaz pe yeh table khud banao, bina dekhe: A aur B ke chaar jod (00, 01, 10, 11), aur har jod ke liye AND, OR, aur “alag hain?” ka jawab. Aakhri column khud sochna padega. Jab ban jaaye, aapne aaj gate design kiya.

SOCHNE KE LIYE

1. (derivation) NOT gate hata do. Sirf AND aur OR bache. Kya “alag hain?” wala purza ab bhi ban sakta hai? Kyun ya kyun nahi?

Jawab: Nahi ban sakta. AND aur OR dono “milaane” wale hain: agar saare inputs 1 hain, toh dono ka output 1 hi hoga, ulatne wala koi nahi. Lekin “alag hain?” ko 11 pe 0 dena hota hai. Bina kisi ulatne wale purze ke, 1 se 0 banta hi nahi. Isliye NOT zaroori hai. Sabak: chhota sa purza bhi poore system ki taakat badal deta hai, aur design ka matlab hi yeh jaan-na hai ki kaunsa purza kya jodta hai.

Chapter 1.6 [SPINE]: Ek machine sab kuch kyun kar sakti hai

Ek ajeeb baat pe kabhi dhyaan diya? Mixer sirf peesta hai. Fridge sirf thanda karta hai. Car sirf chalti hai. Har machine EK kaam karti hai.

Lekin aapka phone? Subah alarm, phir camera, phir bank, phir game, phir video call. Ek hi machine, hazaar kaam. Duniya ki baaki har machine se alag. Aisa kya hai isme?

Chapter 1.3 se recipe wali baat uthao. Machine recipe chalati hai. Ab ek sawal jo poori history ka sabse sundar sawal hai:

Recipe khud kya hai?

Recipe kadmon ki list hai: “jodo,” “compare karo,” “agar aisa toh wahan jao.” Ab in kadmon ko number de do: jodo = 1, compare = 2, copy = 3... Toh poori recipe ban gayi numbers ki qataar: 1, 3, 2, 7...

Ruko. Numbers ki qataar? Woh toh machine switchon mein rakh sakti hai (Chapter 1.4). Matlab:

Recipe usi jagah rakhi ja sakti hai jahan data rakha jaata hai. Aur machine ko aisa banao ki woh apni memory se agla number padhe, aur us number wala kaam kare. Bas.

Iska matlab pheliye, kyunki yeh Level 3 ka janam hai:

Naya kaam chahiye? Nayi machine MAT banao. Bas memory mein nayi recipe daalo. Machine wahi, kaam naya. Camera app dala toh phone camera ban gaya, bank app dala toh bank ban gaya. Machine badli nahi, sirf numbers badle.

Aur ab paisa dekho. Machine banana: factory, saamaan, mazdoor, har copy ka kharcha. Recipe banana: ek baar dimaag lagao, phir har copy LAGBHAG MUFT, kyunki copy sirf numbers ki nakal hai. Chapter 0.2 ka leverage dial yaad karo. Software woh pehli cheez hai jahan leverage ka dial theoretical had tak ghoomta hai: ek baar likho, arab baar chalao, copy ka daam zero.

Isi ek idea ne duniya ki ameer ki naksha badla. Microsoft, Google, Infosys, sab is ek baat pe khade hain: recipe number hai, number ki copy muft hai.

NAAM

Is design ka naam **stored-program computer** hai, aur idea jis se juda hai woh naam hai **von Neumann architecture** (1945). “Ek machine jo koi bhi recipe chala sakti hai” ke idea ko **universal machine** kehte hain, yeh Alan Turing ne 1936 mein, machine banne se pehle, kaagaz pe saabit kiya tha. Aapke phone ka formal naam yahi hai: universal machine.

ASLI DUNIYA SE EK EXAMPLE

Nokia duniya ki sabse badi phone company thi. Uske phone EK kaam ke liye bane the: call aur SMS. iPhone aaya aur usne phone ko universal machine bana diya: app store matlab “koi bhi recipe daalo.” Nokia hardware behtar banata raha, Apple recipes ka bazaar le gaya. Dus saal mein Nokia khatam. Machine vs universal machine ki ladai mein universal hamesha jeetti hai, kyunki uske paas duniya bhar ke recipe likhne walon ki fauj hai.

YAHAN LOG KYA GALAT SAMAJHTE HAIN

“Software company banane ke liye factory jaisa bada paisa chahiye.” Ulta. Software ki factory hai hi nahi, isliye shuru karne ka kharcha history mein sabse kam hai: ek laptop, ek dimaag. Isi liye do do bachche garage mein Google shuru kar paaye. Aur isi liye competition bhi bhayankar hai: jo cheez aap sasta shuru kar sakte ho, woh koi bhi sasta shuru kar sakta hai. Scarcity ab recipe ke IDEA aur QUALITY mein hai, shuru karne mein nahi.

MAP PE

Company case: Zerodha. Do bhaiyon ne stock broking ki recipe software mein likhi. Na branch kholi, na agents rakhe. Ek recipe, ek crore se zyada users, aur India ki sabse zyada munafa wali startups mein ek, bina bahar ka paisa liye. Universal machine pe

likhi ek achhi recipe kya kar sakti hai, iska desi saboot. Aapka Level 4 wala plan isi zameen pe khada hoga.

KHUD DEKHO (5 minute)

Apne phone ki settings mein jao, storage dekho. Wahan apps ki list hogi, har ek ke aage MB/GB. Ab socho: yeh har entry ek recipe hai, numbers ki qataar, jo isi ek machine ko sau alag machines banati hai. WhatsApp ~100 MB ki recipe hai. Us recipe ki har copy muft banti hai, aur 300 crore log chalate hain. Ab formula khud lagao: size x scarcity x leverage.

SOCHNE KE LIYE

1. (derivation) Recipe data ki jagah rehti hai. Toh ek recipe doosri recipe ko PADH ya LIKH bhi sakti hai, kyunki woh bas numbers hai. Is baat se ek bahut badi cheez possible hoti hai jo aap roz use karte ho. Kya?

Jawab: Update. Jab WhatsApp khud ko update karta hai, toh ek recipe (updater) doosri recipe (purana WhatsApp) ko naye numbers se badal rahi hoti hai. Machine ko haath lagaye bina, duniya bhar ki machines raat-o-raat naya kaam seekh jaati hain. Aur isi darwaaze se virus bhi ghusta hai: woh bhi ek recipe hai jo doosri recipes ko likhne ki koshish karti hai. Ek hi taakat, achhe aur bure dono haath mein. Part 3 mein iski hifaazat milegi.

SECTION 2: NUMBERS AUR SIZE

Paanch chapters. Section 1 ne dikhaya ki machine numbers pe recipe chalati hai. Ab do sawal bache: duniya ki cheezein (text, photo, awaaz) numbers kaise banti hain, aur numbers kitni jagah lete hain.

Yeh aapke business decision mein kahan aayegi: Har tech product ka kharcha teen cheezon se banta hai: data kitna bada hai, kitna tez chahiye, kitna rakhna hai. Yeh section aapko woh naap deta hai. Jis founder ko GB aur millisecond ka matlab pata hai, woh engineer ke estimate pe sawal utha sakta hai aur cloud ke bill ko padh sakta hai. Jo nahi jaanta, woh andhere mein sign karta hai.

Chapter 2.1 [SPINE]: Ek cheez kitni jagah leti hai

Kapda meter mein napta hai, doodh litre mein, aur digital cheezein? Aapne naam sune hain: MB, GB. Ab unhe kamaane layak samajh lete hain.

Naap ki ikai wahi hai jo Chapter 1.4 mein bani: **bit**, ek switch ki haalat. Aath bit ka jhund **byte**. Ek byte mein 256 haalat aati hain, itna kaafi hai ek akshar rakhne ke liye. Toh ek moti baat pakdo:

1 byte = lagbhag 1 akshar.

Ab seedhi ginti, har kadam pe hazaar ka guna:

1 KB (kilobyte)	= ~1 hazaar byte	= ek paragraph
1 MB (megabyte)	= ~10 lakh byte	= ek moti kitaab ka text
1 GB (gigabyte)	= ~100 crore byte	= 1,000 moti kitaabein
1 TB (terabyte)	= ~1 lakh crore byte	= ek chhoti library

Ab asli duniya ke anchors, yeh ratne laayak hain kyunki har baat-cheet mein aayenge:

WhatsApp message (sirf text)	~ 100 byte
Ek photo (phone camera)	~ 3 MB
Ek gaana	~ 5 MB
Ek ghanta video call	~ 500 MB - 1 GB
Ek movie (achhi quality)	~ 2-4 GB
Aapke phone ka poora storage	~ 128-256 GB

Ek pattern dikh raha hai? Text nanha hai. Photo usse hazaar guna. Video photo se bhi sau guna. **Text : photo : video = 1 : hazaar : lakh**. Yeh anupaat is poori kitaab ka sabse kaam ka number hai, kyunki jahan size hai wahan kharcha hai.

Kyun aisa? Text mein har akshar ek byte hai. Photo mein har bindu (pixel) ke teen byte hain, aur bindu lakhs mein hain. Video mein har second pe 30 photos hain. Ginti khud guna hoti jaati hai.

NAAM

In naapon ka poora parivaar **storage units** kehlata hai. Ek baareek baat: internet ki speed **bits** mein napti hai (Mbps = megabits per second), storage **bytes** mein (MB). Bit byte ka aathvan hissa hai. Isliye “100 Mbps” ka connection ek second mein sirf 12.5 MB laata hai. Telecom companies bits mein isliye bolti hain kyunki number aath guna bada dikhta hai. Ab aap yeh chaal pakad sakte ho.

ASLI DUNIYA SE EK EXAMPLE

WhatsApp par bheji photo halki kyun ho jaati hai? Kyunki WhatsApp use bhejne se pehle daba deta hai (compress): 3 MB se ~100 KB, tees guna chhoti. Karodo photos roz jaati hain; agar full size jaati, toh WhatsApp ka data kharcha tees guna hota aur aapka data pack bhi. Ek compression ki recipe ne company ke arabon bachaye. Size ka intezaam hi paisa hai.

YAHAN LOG KYA GALAT SAMAJHTE HAIN

Log KB, MB, GB ko bas “chhota, bada, aur bada” samajhte hain. Nahi: har kadam HAZAAR ka guna hai. MB aur GB ka farq wahi hai jo 1 rupaye aur 1,000 rupaye ka. Jab engineer kahe “yeh file GB mein hai,” toh woh keh raha hai “yeh MB wali se hazaar guna bhaari hai,” aur uska kharcha bhi waisa hi hoga.

MAP PE

Rupaye ka rasta: aap Google ko storage ka kiraya dete ho (Google One: 100 GB ke liye mahina ~130 rupaye). Google woh data jinke machines pe rakhta hai, unki disks Samsung/Seagate se aati hain. Neeche tak jao toh ek zaroorat milegi: “meri yaadein na khoyein” (Level 1: yaadon ki hifaazat). Us zaroorat ka poora stack: photos app (L2 service), cloud storage (L3), disk factory (L3 ka hardware kona). Har GB kiraya de raha hai.

KHUD DEKHO (5 minute)

Phone mein koi ek photo kholo, uski details/info dekho (size milega, ~2-4 MB). Ab ek text message ka size socho (~100 byte). Ab khud gino: ek photo kitne messages ke barabar hai? (~30,000). Ab samjho ki data pack video dekhne se hi kyun khatam hota hai, text se kabhi nahi.

SOCHNE KE LIYE

1. (derivation) Do startup ideas: ek mein log 200-word reviews likhte hain, doosre mein log 1-minute video reviews banate hain. Dono ke 10 lakh users hain. Sirf size ke gyaan se batao: kiska kharcha kitna zyada hoga, aur iska business pe kya asar padega?

Jawab: Text review ~1 KB, video review ~50-100 MB: farq lagbhag LAKH guna. Toh video wale ka storage aur bhejne ka kharcha bhi hazaaron-lakhon guna hoga. Asar: text wala app muft users pe bhi zinda reh sakta hai, video wale ko pehle din se paise ka raasta chahiye (ads, subscription), warna har naya user use gareeb karta hai. Product ki shakal uske business model ki shakal tay kar deti hai. Yeh formula ab aapke paas hai, kisi bhi idea pe lagao.

Chapter 2.2 [SPINE]: Text, photo, awaaz, video: sab numbers hain

Machine sirf numbers jaanti hai. Toh aapki shaadi ki photo, maa ki awaaz wali recording, yeh kitaab ka text, yeh sab machine mein kaise ghusa? Har ek ka jawab ek hi chaal se niklega. Chaal yeh hai:

Cheez ko chhote chhote tukdon mein kaato, har tukde ko ek number do, aur numbers ka matlab pehle se sab mil kar tay kar lo.

Ab chaaron pe lagao:

Text. Tukda = akshar. Sab ne mil kar tay kiya: A = 65, B = 66, space = 32. Toh "HI" machine mein 72, 73 hai. Bas. Aapke naam ka spelling numbers ki qataar hai. (Hindi ke aksharon ke bhi numbers tay hain, अ = 2309.)

Photo. Tukda = bindu (pixel). Photo ko jaali mein kaato, har khaane ka ek rang. Aur rang? Har rang laal, hara, neela ke mel se banta hai, toh teen numbers (0-255) har bindu ke liye: (255, 0, 0) = poora laal. 40 lakh bindu x 3 numbers = aapki photo.

Awaaz. Awaaz hawa ka kampan hai. Tukda = waqt ka pal. Har second ko ~44,000 palon mein kaato, har pal pe kampan ki oonchai naapo, woh number likh lo. Numbers ki qataar wapas bajao, awaaz wapas. Maa ki awaaz 44,000 numbers prati second hai.

Video. Tukda = frame. Har second 30 photos, plus awaaz. Isliye video sabse bhaari hai: photo ka kharcha x 30 x seconds.

Teeno kadmon mein sabse zaroori kaunsa hai? Kaatna? Naapna? Nahi: **"matlab pehle se sab mil kar tay kar lo."** 72 ka matlab H tabhi hai jab bhejne wala aur paane wala dono ek hi table maante hon. Agar mera 72 H hai aur tumhara kuch aur, toh message kachra ban jaayega. Yeh baat, ki **numbers ka matlab samjhauta hai, numbers mein nahi hota**, aage internet (Part 3) aur AI (Book 2) dono ki neev hai.

NAAM

Matlab tay karne wale table ka naam **encoding** hai. Text ka sabse purana table **ASCII** (1963), aaj ka poora table **Unicode**, jisme duniya ki har bhasha aur emoji tak hai (emoji bhi bas ek number hai: “laughing face” = 128514). Photo ke bindu ka naam **pixel**, awaaz ke pal ka naam **sample**, video ki photo ka naam **frame**.

ASLI DUNIYA SE EK EXAMPLE

Jab aap Hindi mein type karte ho aur kisi purani website pe ??? ya dibbe dikhte hain, toh kya hua? Aapke numbers pahunch gaye, lekin us taraf ka software Unicode ka Hindi wala hissa nahi jaanta tha. Numbers sahi, matlab kho gaya. Poori duniya ka software dheere dheere Unicode pe isliye aaya kyunki bina ek jaisi table ke, bhaashaon ka internet ban hi nahi sakta tha. Ek boring sa samjhauta, arabon logon ko jodne ki shart nikla.

YAHAN LOG KYA GALAT SAMAJHTE HAIN

Log sochte hain machine photo “dekhti” hai ya gaana “sunti” hai. Nahi. Machine ke liye photo 1.2 crore numbers hain aur gaana 1.5 crore. Usko na rang pata hai na raag. Aur phir bhi (yeh aage Book 2 ka beej hai) numbers ke pattern se machine bata deti hai ki photo mein billi hai. Dekhna nahi, ginti se pehchaan. Jab woh baat aaye, yeh chapter yaad rakhna: sab kuch numbers hi tha, hamesha.

MAP PE

Company case: JioCinema/Hotstar jab cricket dikhata hai, toh 4 crore log ek saath frames ke numbers maang rahe hote hain. Unka poora engineering kharcha isi chapter ke anupaat se aata hai: video = sabse bhaari numbers. Aur unka business model bhi wahi tay karta hai: ads aur subscription zaroori hain, kyunki har viewer ka data kharcha asli hai. Level 1 zaroorat: manoranjan. Uske numbers: sabse mehnges.

KHUD DEKHO (5 minute)

Kisi ko yeh message bhejo: “72 69 76 76 79”. Aur bolo har number 64 ghata ke $A=1$, $B=2$ wali ginti pe akshar banao. Jawab aayega HELLO. Aapne abhi apna encoding table banaya aur ek insaan ko decode karwaya. Machine bas yahi karti hai, arab guna tez.

SOCHNE KE LIYE

1. (derivation) WhatsApp voice message bhejte waqt “bhejne se pehle daba do” (compress) kaise possible hai? Awaaz toh naapi hui thi, 44,000 number prati second. Kaunse numbers phenke ja sakte hain?

Jawab: Woh jo insaan ke kaan waise bhi nahi pakadte. Bahut ooncha kampan, bahut halki awaaz jo tez awaaz ke peeche dab gayi, do palon ke beech ka farak jo andaaze se wapas ban sakta hai. Recipe unhe phenk deti hai, size das guna girta hai, kaan ko farq nahi padta. Sabak: compression ka matlab hai “matlab bachao, numbers phenko.” Aur dhyaan do, yeh ek FAISLA hai ki kya phenkna hai, aur har faisla kisi insaan ki recipe mein likha hai. Machine mein har jagah aise chhupe hue faisle milenge.

Chapter 2.3 [SPINE]: Bade numbers ka matlab

Is kitaab mein ab tak aaye: 15 arab transistor, 44,000 sample prati second, arab kadam prati second. Aage aur bade aayenge. Problem yeh hai ki insaan ka dimaag lakh ke aage sab numbers ko ek jaisa “bahut bada” maan leta hai. Woh chalega nahi, kyunki in numbers ke beech ke FARQ mein hi saari samajh aur saara paisa hai.

Pehle tarjuma ka table, kyunki India ginti apni bolta hai aur technology angrezi:

10 lakh	= 1 million	(M)	
1 crore	= 10 million		
1 arab	= 1 billion	(B)	= 100 crore
1 kharab	= 100 billion		
	= 0.1 trillion	(T)	

Ab farq ko mehsoos karo, ghadi se. Ek second mein ek ginti bolo:

1 million tak ginne mein	~ 11 din lagenge
1 billion tak	~ 31 SAAL
1 trillion tak	~ 31,000 saal

Million aur billion “paas paas” nahi hain. Unke beech hazaar guna ka samundar hai. Jab news kahe “company X billion dollar ki hui,” toh ab aapke paas paimana hai.

Ab ulti taraf, chhote waqt ki ginti, kyunki machine wahan jeeti hai:

1 second	aapka ek palak jhapakna
1 millisecond	(1/1000 sec) achhe internet ka ek phera
1 microsecond	(1/10 lakh) machine memory se ek cheez uthana
1 nanosecond	(1/1 arab) CPU ka ek kadam

Ek moti baat nikaalo: **CPU ke ek kadam (nanosecond) aur internet ke ek phere (millisecond) mein 10 lakh guna ka farq hai.** Matlab jab tak ek message shehar paar jaa kar lautta hai, CPU 10 lakh kadam chal chuka hota hai. Machine ke andar sab kuch

muft jaisa tez hai; bahar jaana mehnga hai. Yeh ek line aage Part 3, 4, 5 mein baar baar kaam aayegi, aur engineers ke har design ke peeche yahi hai: **bahar mat jao jab tak zaroori na ho.**

NAAM

In chhote naapon ke naam upar table mein hain: milli (hazaarvan), micro (10 lakhvan), nano (arabvan). Speed ke is khel ka naam **latency** hai: ek kaam shuru hone se jawab aane tak ka waqt. Yeh shabd Part 5 mein poora chapter layega.

ASLI DUNIYA SE EK EXAMPLE

Google ne naapa ki search 400 millisecond dheemi hui toh log kam search karne lage. Aadha palak jhapakne se kam waqt, aur karodon ka farq. Amazon ki ginti: har 100 millisecond ki deri = ~1% kam bikri. Isliye speed engineering ki shauk nahi, seedha paisa hai. Jo company millisecond gin sakti hai, woh unhe रुपये mein badal sakti hai.

YAHAN LOG KYA GALAT SAMAJHTE HAIN

”Computer tez hai, bas.” Nahi: computer KAHAN tez hai, yeh jaanna hi asli gyaan hai. Andar nanosecond, bahar millisecond, 10 lakh ka farq. Jo yeh nahi jaanta, woh aisa product sochta hai jo har kadam pe internet se poochhta hai, aur phir hairaan hota hai ki sab dheema kyun hai. Tez machine pe bhi dheema design possible hai, aur aam hai.

MAP PE

Kaun kamata hai is chapter se: jo scale ko padh sakta hai. Ek investor ke liye “10 lakh users” aur “1 crore users” mein das guna ka farq hai; jo founder dono ko “bahut saare users” bolta hai, usse paisa door rehta hai. Aur jo engineer millisecond bacha sakta hai, woh Google/Amazon jaise business mein seedha revenue dial ghuma raha hai, isliye aisi skill ki tankhwah crore paar jaati hai. Numbers ko mehsoos karna Level 2, 3, 4 teeno ki shared bhasha hai.

KHUD DEKHO (5 minute)

Phone pe stopwatch chalao aur 100 tak zor se gino. (~100 second.) Ab hisaab: 1 million tak aise ginne mein kitne din? (~11 din, bina soye.) 1 billion tak? (~31 saal.) Ek baar khud hisaab lagaoge toh billion shabd hamesha ke liye bhaari ho jaayega, jaisa use hona chahiye.

SOCHNE KE LIYE

1. (derivation) Aapke phone ki chip 15 arab transistor ki hai. Agar har transistor chawal ka ek dana hota, toh kitna chawal banta? (Ek kilo mein ~50,000 dane.) Aur is se kya samajh aata hai ki chip banane wali factory duniya mein teen-chaar hi kyun hain?

Jawab: $15 \text{ arab} / 50,000 = 3 \text{ lakh kilo}$, yaani 300 tonne chawal, pachees trucks bhar ke. Itni cheezein nakhoon bhar jagah pe, bina ek bhi galat lage, chhaapni hain. Isliye yeh duniya ki sabse mushkil manufacturing hai: ek factory lagane ka kharcha ~2 lakh crore रुपये, aur hunar sirf TSMC (Taiwan), Samsung (Korea), Intel (US) ke paas. Scarcity itni oonchi ki desh iske liye jang ki taiyari karte hain. Ab aap samajhte ho ki “chip war” news mein kyun rehti hai.

Chapter 2.4 [DEPTH]: Ek instruction chalti kaise hai

(DEPTH chapter. Skip kar sakte ho, lekin yeh chhota hai aur iske baad “processor” shabd hamesha ke liye khul jaata hai.)

Chapter 1.6 se: recipe numbers ban kar memory mein rakhi hai. Machine use chalati kaise hai? Poora raaz ek ginne wale khaane mein hai.

CPU ke andar ek chhota sa khaana hai jisme bas ek number rehta hai: **”abhi main recipe ke kadam number ____ pe hoon.”** Aur CPU din raat bas yeh teen kaam gol gol karta hai:

1. UTHAO (fetch): khaane mein jo number hai, memory ki us jagah se kadam utha lo. Maan lo wahan likha hai: “3” (matlab: jodo).

2. SAMJHO (decode): number 3 ka matlab kya hai? Chapter 1.5 wali gates ki chain number ko padh kar sahi purza chuun leti hai: jodne wala dibba on.

3. KARO (execute): jod do. Jawab kisi khaane mein rakho. Aur ginne wale khaane mein 1 badha do, taaki agla phera agla kadam uthaye.

Bas. Yeh teen kadam ka chakkar hi “computer chal raha hai” ka poora matlab hai. Aapke phone mein yeh chakkar ek second mein ~3 arab baar ghoomta hai.

Ek sawal khud se poochho: “agar aisa toh wahan jao” wala kadam kaise chalega? Aasaan: woh kadam bas ginne wale khaane mein NAYA number likh deta hai. Ab agla fetch wahin se uthega. Recipe ne chhalaang maar li. Loop bhi yahi hai: peechhe wale number pe chhalaang, baar baar. Programming ka poora control isi ek khaane se hota hai.

Ab “3 GHz processor” ka matlab bhi le lo: GHz = giga hertz = arab phere prati second. 3 GHz matlab yeh chakkar 3 arab baar prati second. Yeh hi woh “arab kadam” hai jo kitaab mein baar baar aaya.

NAAM

Ginne wale khaane ka naam **program counter** hai. Teen kadam ke chakkar ka naam **fetch-decode-execute cycle**. Chakkar ki raftaar ka naam **clock speed** (GHz mein). Kadmon ke numbers wali poori bhasha ka naam **machine code**: yehi ek bhasha hai jo CPU sach mein samajhta hai. (Insaan iski jagah aaraam wali bhaashaein likhta hai; unka tarjuma Part 2 ka mukhya kissa hai.)

ASLI DUNIYA SE EK EXAMPLE

Phone kabhi “hang” hota hai. Ab aap bata sakte ho andar kya hua: program counter kisi aisi jagah phas gaya jahan recipe gol gol ghoom rahi hai aur bahar aane ki shart kabhi sach nahi ho rahi (infinite loop), ya kisi aise number pe chala gaya jahan recipe thi hi nahi. Machine kharab nahi hui, recipe ne use gadhe mein ghuma diya. Restart kya karta hai? Program counter samet sab kuch shuruaati jagah pe wapas. Isliye restart itni beemariyan theek karta hai.

YAHAN LOG KYA GALAT SAMAJHTE HAIN

”Zyada GHz = tez computer,” bas itna. Adhoora hai. GHz sirf chakkar ki raftaar hai; kaam kitna hua yeh is pe bhi hai ki har phere mein kitna hota hai, kadam kitne samajhdaar hain, aur memory kitni tez kadam pakda rahi hai (agla chapter). Do processor same GHz pe do guna alag kaam kar sakte hain. Dukaandaar aapko ek number bech raha hai, machine kai numbers ka khel hai.

MAP PE

Rupaye ka rasta: Apple apne chips khud design karta hai (M-series) taaki har phere mein zyada kaam ho. Behtar cycle = kam battery mein zyada kaam = mehngi machine bikti hai. Ek chakkar jo aapne abhi 10 minute mein samjha, uski bareeki pe duniya ki sabse keemti company ka margin tika hai. Level 3 ke andar bhi levels hain, aur neeche wala jitna behtar, upar sab utna behtar.

KHUD DEKHO (5 minute)

Yeh recipe kaagaz pe chalao, aap khud CPU bano. Khaana K mein 5 hai. Recipe:
kadam 1: K mein 3 jodo. kadam 2: agar $K < 15$, kadam 1 pe jao. kadam 3: ruko. Ungli se program counter banao aur chalao. Kitne phere lage? (K: 5, 8, 11, 14, 17 → 4 jod, phir shart fail, ruk gaye.) Aapne abhi fetch-decode-execute aur loop dono chala liye.

SOCHNE KE LIYE

1. (derivation) Program counter mein galat number aa jaaye, aisi jagah ka jahan recipe nahi, kachra numbers hain. CPU kya karega? Usko kaise pata chalega ki yeh “recipe nahi” hai?

Jawab: Nahi pata chalega, yahi darr ki baat hai. CPU ke liye har number kadam jaisa hi dikhta hai, woh kachre ko bhi decode karke chalane ki koshish karega. Kabhi bekaar kaam hoga, kabhi crash. Isliye aas paas ke intezaam (operating system, Part 2) deewarein banate hain: kaunsi recipe kahan chal sakti hai, kaunsi memory chhoo sakti hai. Aur hackers ka aadha khel hi yeh hai: kisi tarah apne numbers program counter tak pahunchana. Ek chhota khaana, poori security ki jang ka maidan.

Chapter 2.5 [SPINE]: Yaad rakhna aur bhool jaana

Ek paheli se shuru karo. Phone pe game khel rahe the, battery achanak khatam. Charge karke on kiya: game wahin ka wahin nahi hai, dobara kholna pada, lekin aapki photos sab salaamat hain. Ek hi machine mein kuch cheezein bijli jaane se mit gayin, kuch nahi mitin. Kyun?

Kyunki machine mein yaadaasht EK nahi, DO hain, aur dono ka sauda alag hai:

Tez wali, bhulakkad. Switchon se bani yaadaasht (Chapter 1.4 wali). Bijli hai toh haalat hai; bijli gayi, sab 0. Lekin raftaar? CPU ke kadmon ke saath bhaag sakti hai. Yahan woh sab rehta hai jo ABHI chal raha hai: khula hua app, game ki abhi wali haalat.

Dheemi wali, pakki. Aisi jagah jahan bits bijli ke bina bhi tike rehte hain (chumbak ya phanse hue charge se). Bijli jaao ya aao, data wahin. Lekin dheemi: tez wali se sauhazaar guna. Yahan photos, apps, files rehti hain.

Ab poochho: dono ke faayde wali ek cheez kyun nahi bana lete, tez bhi, pakki bhi, sasti bhi? Engineers 70 saal se koshish kar rahe hain. Physics ka sauda hai: **tez, sasta, pakka: teeno mein se do chun lo.** Isliye machine dono rakhti hai aur beech mein data ka aana jaana chalta rehta hai:

App “kholna” = pakki wali se tez wali mein recipe copy karna. Isliye kholne mein waqt lagta hai. File “save karna” = tez wali se pakki wali mein likhna. Isliye save ka matlab hai “ab bijli jaaye toh bhi bacha rahega.” Phone “hang by memory” = tez wali bhar gayi, ab machine har kadam pe dheemi wali se kaam chala rahi hai.

NAAM

Tez bhulakkad ka naam **RAM** (ya bas memory). Dheemi pakki ka naam **storage** (disk, SSD). Dibbe pe “8 GB RAM, 128 GB storage” ka matlab ab poora khulta hai: 8 GB ka tez kaam-karne-ka mez, 128 GB ki pakki almaari. Mez chhota aur mehnga, almaari badi aur sasti. Yehi anupaat har machine mein milega, phone se datacenter tak.

ASLI DUNIYA SE EK EXAMPLE

Sasta phone “12 GB storage full” nahi bolta, woh chalte chalte atakta hai: wajah RAM kam hai, mez chhota hai, har app switch pe almaari ke chakkar lag rahe hain. Dukaan wala aapko storage ka bada number dikhata hai (128 GB!), RAM chhupa deta hai (4 GB). Ab aap dibba padh kar asli sawal pooch sakte ho: mez kitna bada hai? Yeh chhota sa gyaan hazaaron rupaye ki galat khareed se bachata hai.

YAHAN LOG KYA GALAT SAMAJHTE HAIN

”Memory full ho gayi, purani photos delete karo.” Do alag cheezein mix ho gayin: photos storage mein hain, hang hona RAM ka hai. Photos hataane se hang theek nahi hota (jab tak storage bilkul hi thasa na ho). Aur ulta bhi: “8 GB RAM hai toh 8 GB ki file aaraam se khulegi” bhi seedha nahi hai, mez pe aur cheezein bhi rakhi hain. Do naam, do kaam, ab hamesha alag rakhna.

MAP PE

Company case: memory chips ka bazaar (Samsung, SK Hynix, Micron) ~150 arab dollar ka hai, aur AI ne use aag laga di hai, kyunki AI recipes ka mez (RAM) rakshasi size ka chahiye. 2024-25 mein AI wale memory chips (HBM naam ke) ki itni kami hui ki daam aasman pe chale gaye. Chapter 0.2 ka formula: scarcity ka dial. Jab bhi koi nayi technology aati hai, uske neech wale kisi purze ki scarcity kisi ka sona ban jaati hai. Yeh pattern pehchaanna hi mauka pehchaanna hai.

KHUD DEKHO (5 minute)

Koi bhaari app kholo (game ya editor): kitne second lage? Yeh pakki almaari se tez mez tak copy ka waqt tha. Ab use band karke TURANT dobara kholo: is baar tez khula na? Kyunki system ne use mez ke paas hi rakha tha. Aapne abhi dono yaadaashton ka farq apni ghadi se naap liya.

SOCHNE KE LIYE

1. (derivation) Ek cheez ka jawab is chapter se nikaalo: “save” button dheere dheere duniya se gayab kyun ho raha hai? (Google Docs mein save button nahi hai.)

Jawab: Pehle save isliye tha ki tez wali yaadaasht se pakki wali mein likhna mehnga kaam tha, toh machine aapse poochhti thi kab likhna hai. Ab storage tez aur sasta ho gaya, toh software har kuch second mein chupchaap khud likh deta hai (auto-save), aur internet pe bhi copy bhej deta hai. Jab kisi kaam ka kharcha gir jaata hai, toh us se juda hua button bhi mar jaata hai. Design ko technology ki economics chalati hai, yeh baat product banate waqt hamesha saath rakhna.

PART 1 KA ANT: NAKSHA AUR AGLA KADAM

Jo aapke paas ab hai

PAISA	bharose ka intezaam, sirf Level 1 pe ghusta hai
KAMAAI KA FORMULA	size x scarcity x leverage (ghante nahi)
CHAAR LEVELS	zaroorat -> business -> technology -> AI paisa neech se upar behta hai
ABSTRACTION	har level neech wale ko chhupa deta hai, aur chhupa hissa hi mehnga hai
MACHINE	arabon switch (transistors), designed chains mein
BIT / BYTE	switch ki haalat / 8 switchon ka jhund
GATES	AND, OR, NOT: inhi se jod, compare, "agar"
COMPUTER	numbers pe pehle se likhi recipe, arab kadam prati second, bina thake
UNIVERSAL MACHINE	recipe bhi number hai, isliye ek machine sab kuch: yahi software ka janam
SAB KUCH NUMBER	text/photo/awaaz/video: kaato, naapo, matlab pehle se tay karo (encoding)
NAAP	KB->MB->GB->TB, har kadam hazaar guna; text : photo : video = 1 : hazaar : lakh
ANDAR VS BAHAR	CPU ka kadam nanosecond, internet ka phera millisecond: 10 lakh guna farq
DO YAADAASHT	RAM (tez, bhulakkad) vs storage (dheemi, pakki)

Ek test, khud ke liye

Bina kitaab palte, kisi ko (ya khud ko, bol kar) yeh 4 jawab do:

1. Paisa sirf Level 1 pe kyun ghusta hai?
2. Surgeon se zyada app wala kyun kamata hai?
3. Ek hi phone camera bhi, bank bhi, game bhi kaise ban jaata hai?
4. "8 GB RAM, 128 GB storage" ka matlab kya hai?

Agar chaaron aa gaye, Part 2 ke liye tayyar ho. Jo atka, uska chapter number upar wale naksha se milega, wahin lauto.

Part 2 mein kya hai

Machine recipe chalati hai, yeh aapko pata chal gaya. Ab agla sawal: recipe LIKHI kaise jaati hai? Insaan ki bhasha se machine ke numbers tak ka tarjuma kaun karta hai?

Languages itni saari kyun hain? Woh manager kaun hai jo saari recipes ko ek machine pe ladne se rokta hai (operating system)? Aur software mein bug hote hi kyun hain?

Part 2: MACHINE SE BAAT.

PART 1 KA MINI-GLOSSARY

Sirf woh shabd jo is part mein aaye. Har ek ke aage uska chapter, taaki lautna aasan ho.

abstraction	andar ki complexity chhupa kar seedha handle dena (0.4)
AND / OR / NOT	teen buniyaadi gates: dono/koi-bhi/ulta (1.5)
ASCII / Unicode	aksharon ke numbers ka samjhauta, purana / aaj ka (2.2)
binary	do-guna wali ginti, sirf 0 aur 1 se (1.4)
bit	ek switch ki haalat: 0 ya 1 (1.4)
byte	8 bits; lagbhag ek akshar (1.4)
circuit	bijli ka bandh raasta (1.1)
clock speed	CPU ke chakkar ki raftaar, GHz mein (2.4)
CPU / processor	recipe chalane wala mukhya purza (1.3)
double coincidence of wants	dono ko ek hi waqt ek doosre ki cheez chahiye (0.1)
encoding	numbers ka matlab tay karne wala table (2.2)
fetch-decode-execute	CPU ka teen-kadam chakkar: uthao, samjho, karo (2.4)
frame	video ki ek photo; ~30 prati second (2.2)
hardware	chhoone laayak machine (1.3)
KB / MB / GB / TB	size ki seedhi, har kadam hazaar guna (2.1)
latency	kaam shuru se jawab tak ka waqt (2.3)
ledger	len-den ka register; paisa chalta-phirta ledger hai (0.1)
leverage	kaam ek baar, faayda baar baar (0.2)
logic gates	switchon se bane decide karne wale purze (1.5)
machine code	numbers wali bhasha jo CPU seedha samajhta hai (2.4)
market size	problem kitne logon ki, kitni gehri (0.2)
pixel	photo ka ek bindu; teen numbers ka rang (2.2)
program	machine ke liye likhi recipe (1.3)
program counter	"abhi kadam number __ pe hoon" wala khaana (2.4)
RAM	tez, bhulakkad yaadaasht; kaam ka mez (2.5)
sample	awaaz ka ek pal ka naap; ~44,000 prati second (2.2)
scarcity	kitne kam log yeh kar sakte hain (0.2)
software	saari recipes, milkar (1.3)
stack	levels ka minaar: kaun kis pe khada hai (0.3)
storage	dheemi, pakki yaadaasht; almaari (2.5)
stored-program computer	recipe ko data ki jagah rakhne wala design (1.6)
transistor	bijli se dabne wala switch (1.1)
universal machine	ek machine jo koi bhi recipe chala sake (1.6)